

Layer 4 — Application platforms (own the execution model)

Spin / SpinKube
FaaS handler model

wasmCloud
Distributed actors

Slight
WASI-cloud-core

Per-request instantiation — not comparable to long-running Docker containers

Layer 3 — Node provisioning (install shims onto K8s nodes)

KWasm operator
Helm chart, node annotation

Runtime class manager
SpinKube, lifecycle operator

Automates what you can also do manually with SSH + config edits

installs

Layer 2 — Containerd shims (RuntimeClass interface)

runwasi — Rust library, containerd subproject (all shims built on this)

shim-wasmedge-v1

shim-wasmtime-v1

shim-wasmer-v1

embeds

Layer 1 — Wasm runtimes (execute .wasm bytecode)

WasmEdge
CNCF Sandbox

Wasmtime
Bytecode Alliance

Wasmer
Commercial

WAMR
Intel, embedded

wazero
Go library only

wasm3
Interpreter only

wasmi
Rust interpreter

Gray = no containerd shim, no K8s path

Alternative path: crun/youki with Wasm support (OCI low-level runtimes, not shown — different architecture)